

Exploring the Frontiers of Knowledge Distillation: From Basics to Advanced Techniques*

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Abstract

Knowledge Distillation (KD) is a model compression technique where a "student" model learns from a "teacher". This project investigates KD effectiveness beyond standard applications, exploring basic principles (MNIST), temperature tuning, and generalization to unseen classes. We examine advanced paradigms: KD with Quantization-Aware Training (QAT), Reverse KD (small teacher, large student), Multi-Teacher KD (MTKD) using ensembles, and Self-Distillation (SD) for teacher-free improvement. Experiments on MNIST and CIFAR using MLPs/ResNets show KD methods improve accuracy, generalization, and robustness, often acting as effective regularizers. (All experiments were run on NVIDIA L4 GPUs, consuming roughly 60 GPU-hours in total.)

1 Introduction

Knowledge Distillation (KD) [1] addresses the challenge of deploying large, computationally expensive deep neural networks by transferring their knowledge to smaller "student" models. This is often achieved by training the student on the soft probability outputs (soft labels) of the larger "teacher" model, capturing richer "dark knowledge" than hard labels alone.

This project explores KD's versatility. We start with basic KD on MNIST, examining teacher choice, temperature (T) impact, and generalization. We then investigate advanced techniques: (1) Combining KD with QAT; (2) Reverse KD (Re-KD); (3) Multi-Teacher KD (MTKD); (4) Self-Distillation (SD). Our goal is to understand the effectiveness, mechanisms, and application scope of these diverse KD strategies through experiments on MNIST and CIFAR datasets.

2 Related Works

Our work builds on key KD advancements. Hinton et al. [1] introduced the foundational teacher-student framework using soft targets. Yuan et al. [2] proposed teacher-free concepts, highlighted KD's regularization effect, and demonstrated Reverse KD's viability. Zhang et al. [3] developed Self-Distillation where models learn from internal layers. Polino et al. [4] showed KD aids quantized models (QAT). Others [5, ?] explored Multi-Teacher KD using ensembles or adaptive weighting. We experimentally validate several of these paradigms.

*GitLab Repository: <https://github.com/EttaZhouuuu/ECE661Project-KD>

3 Methodology

Standard training practices (SGD, LR schedules) were used with PyTorch.

Basic KD (MNIST): MLP teacher (2x1200) and student (1x400) (Table ??). Compared student trained with hard labels vs. standard KD loss ($L = (1 - \alpha)L_{CE} + \alpha T^2 L_{KL}$). Tuned temperature T . Tested generalization by removing one digit class during student training.

KD with QAT (MNIST): Combined KD with simulated Quantization-Aware Training (INT8, INT4) using STE for gradients. Compared performance against baseline QAT (no KD) and FP32 student.

Reverse KD (CIFAR-10): Used fixed ResNet-34 teacher to train ResNet-50 student via standard KD loss ($T = 5, \alpha = 0.7$).

Multi-Teacher KD (CIFAR-10): Trained ResNet-34 student using averaged soft labels ($T = 5$) from $K = 1..5$ ResNet-50 snapshot teachers (L combination with $\alpha = 0.7$).

Self-Distillation (CIFAR-10): Added auxiliary heads to ResNet-50 (layer2, layer3). Trained using composite loss: $L_{total} = L_{final_CE} + W_{aux} \sum L_{aux_CE} + W_{kd} \sum L_{KD}(T)$. Tuned T, W_{aux}, W_{kd} . Evaluated robustness on CIFAR-10-C.

4 Experiments and Results

4.1 Basic Distillation

The architectures and results on MNIST are summarized in Tables 1 and 2. Soft labels from the larger teacher model notably improved the student’s test accuracy from 98.16% to 98.87%. Although ResNet-18 was tested as a teacher due to its strong image recognition ability, it performed worse than the MLP teacher, likely due to architectural mismatch with the MLP student. Therefore, two MLPs were used for the following experiments.

Table 1: Architecture of teacher and student models

Model	Architecture	FLOPs / Parameters
Teacher Model 1: MLP	(fc1): Linear(784, 1200)	2.4M MACs / 2.4M
	(fc2): Linear(1200, 1200)	
	(fc3): Linear(1200, 10)	
Student Model 1: MLP	(fc1): Linear(784, 400)	318.41K MACs / 318.01K
	(fc2): Linear(400, 10)	
Teacher Model 2: ResNet18	ResNet18 (conv1, 4 residual layers + FC)	457.65M MACs / 11.17M

Table 2: Distillation vs. Baseline Student Performance

Model	Teacher		Student Baseline (MLP)	Distillation	
	MLP	ResNet18		MLP→MLP	ResNet→MLP
Test accuracy (%)	98.96	99.38	98.16	98.87	98.33

In Equation (1):

$$q_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)} \quad (1)$$

where q_i is the softened probability for class i , z_i is the logit for class i , and T is the temperature parameter.

A higher temperature smooths the output probabilities of the teacher, revealing inter-class similarities. This allows the student to learn richer “dark knowledge” and improves generalization compared to training only on hard labels [1].

The effect of temperature T on the distilled student’s test accuracy is summarized in Figure 1. A finer sweep revealed $T = 5$ yielded the highest accuracy.

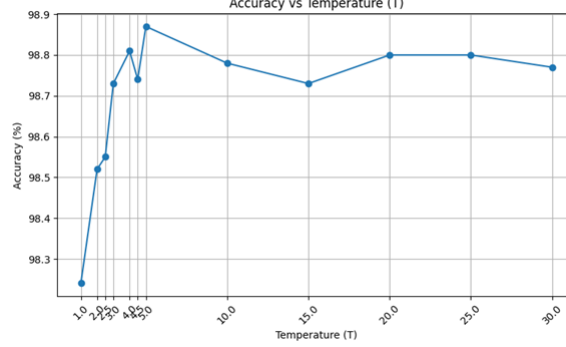


Figure 1: Effect of Temperature on Knowledge Distillation Accuracy

Next, we tested generalization to unseen classes. Removing all ‘3’s or ‘4’s from the training set separately, the distilled student model achieved over 95% accuracy on those digits as shown in Table 3, confirming KD’s ability to transfer knowledge even for unseen classes.

Table 3: Performance on Omitted Digit

Training Method	Training Set	Overall Test Acc (%)	Acc on Digit (%)
Baseline (Cross-Entropy)	MNIST without ‘3’	88.24	0.00
Distillation (T=10)	MNIST without ‘3’	98.56	95.05
Baseline (Cross-Entropy)	MNIST without ‘4’	88.61	0.00
Distillation (T=10)	MNIST without ‘4’	98.64	95.62

4.2 Distillation and Quantization

Table 4: MNIST Student Model Accuracy (Post-Training vs. QAT vs. Quantized Distillation)

Bit Depth	Quantized Test Acc (%)	Accuracy Drop vs FP32 (%)
FP32	98.54	-
QAT Baseline (STE only) INT8	97.98	-0.56
QAT Baseline (STE only) INT4	98.03	-0.51
Quantized Distillation INT8	98.51	-0.03
Quantized Distillation INT4	98.35	-0.19

This work combines Quantization-Aware Training (QAT) with Knowledge Distillation (KD). The student model learns with simulated quantization constraints during training, guided by a full-precision teacher model [4]. Key steps per training iteration:

- **Quantized Forward Pass:** Student’s full-precision weights (w) are temporarily quantized (w_q) using uniform symmetric quantization. The output student_logits reflect quantized performance.
- **Distillation Loss:** Combines KD soft loss and standard hard loss:

$$L = \alpha T^2 \text{KLDiv}(\text{softmax}(\text{student}/T), \text{softmax}(\text{teacher}/T)) + (1 - \alpha) \text{CrossEntropy}(\text{student}, y_{\text{true}})$$

- **STE Backward Pass:** Gradients of L w.r.t. w are computed using the Straight-Through Estimator (STE).

- **Update:** The optimizer updates the student’s full-precision weights.

The results in Table 4 show that standard QAT incurs a $\sim 0.5\%$ accuracy drop. Quantized Distillation significantly outperforms this, nearly matching FP32 accuracy at INT8 and substantially reducing the accuracy loss at INT4.

4.3 Reverse KD (CIFAR-10)

The performance comparison for Reverse KD is shown in Table 5[cite: 274].

- **Baselines:** The ResNet-34 teacher achieved an average accuracy of $88.32\% \pm 0.25\%$, while the larger ResNet-50 student baseline reached $89.02\% \pm 0.23\%$ [cite: 275].
- **Re-KD Results:** When the ResNet-50 student was trained using distillation from the smaller, fixed ResNet-34 teacher, its average accuracy increased to $89.16\% \pm 0.23\%$ [cite: 276]. This result surpassed both the teacher’s and the student’s own baseline performance[cite: 276].
- **Analysis:** This counter-intuitive finding supports the hypothesis by Yuan et al. that KD acts as a powerful regularizer (akin to learned label smoothing)[cite: 271, 278]. The "dark knowledge" or structured output distribution from the smaller teacher helped regularize the larger student, pushing it towards better generalization, regardless of the conventional capacity hierarchy[cite: 278].

Table 5: Reverse Knowledge Distillation Results (CIFAR-10)

Model	ResNet-34 (Teacher)	ResNet-50 (Student Baseline)	Reverse KD (ResNet-34 $\rightarrow$ ResNet-50)
Mean $\pm$ Std Acc (%)	$88.32 \pm 0.25 \%$	$89.02 \pm 0.23 \%$	$89.16 \pm 0.23 \%$

Hyperparameter Sensitivity Analysis: To further understand the interplay between temperature τ and the distillation loss weight α , we performed a grid search over $\tau \in \{1, 5, 10, 20\}$ and $\alpha \in \{0.1, 0.5, 0.9\}$. The results, averaged over multiple runs, are summarized in Table 6.

Table 6: Hyperparameter Sweep for Reverse KD (ResNet-34 $\rightarrow$ ResNet-50, CIFAR-10): Mean Acc $\pm$ Std (%)

Temperature (τ)	$\alpha = 0.1$	$\alpha = 0.5$	$\alpha = 0.9$
1	88.15 ± 0.14	88.63 ± 0.10	88.31 ± 0.22
5	88.20 ± 0.21	88.62 ± 0.18	89.40 ± 0.15
10	88.49 ± 0.19	88.38 ± 0.12	89.10 ± 0.17
20	88.37 ± 0.16	88.96 ± 0.14	88.20 ± 0.20

The sweep revealed that optimal performance ($89.40 \pm 0.15\%$) was achieved at $\tau = 5$ and $\alpha = 0.9$. This suggests that moderate temperature softening ($\tau = 5$) effectively exposes useful dark knowledge without excessive flattening of the distribution, while a high emphasis on the distillation loss ($\alpha = 0.9$) ensures the student closely aligns with the teacher’s softened outputs, maximizing the regularization benefit. Both very low ($\tau = 1$) and very high ($\tau = 20$) temperatures, as well as lower distillation weights ($\alpha \leq 0.5$), resulted in suboptimal performance, underscoring the criticality of calibrating these hyperparameters for effective Reverse KD.

4.4 Multi-Teacher KD (CIFAR-10)

The performance of the ResNet-34 student when distilled from an increasing number of ResNet-50 snapshot teachers is shown in Table 7[cite: 298]. The student baseline accuracy was 95.42

- **Teacher Performance:** The five ResNet-50 snapshot teachers had individual accuracies ranging from 92.52% to 94.85

- **MTKD Results:** Distilling knowledge from an increasing number of teachers (K) progressively improved the ResNet-34 student’s accuracy, from 93.80% ($K=1$) to 95.44% ($K=5$)[cite: 298, 299].
- **Analysis:** Using multiple teachers provided synergistic benefits, allowing the student to surpass the performance of any single teacher when $K \geq 3$ [cite: 300]. The aggregated knowledge from the ensemble provided richer supervision. With all 5 teachers, the student marginally exceeded its own non-distilled baseline performance (95.44% vs 95.42%)[cite: 301].

Table 7: Multi-Teacher Knowledge Distillation Results (ResNet-50 Snapshots $\rightarrow$ ResNet-34 Student on CIFAR-10)

Number of Teachers (K)	1	2	3	4	5
Individual Teacher Accuracies used (%)	92.52	+94.19	+94.48	+94.85	+94.78
Student Accuracy after KD (%)	93.80	94.92	95.10	95.29	95.44

4.5 Self-Distillation (CIFAR-10)

The Self-Distillation experiments compared a baseline ResNet-50 with an SD-enhanced ResNet-50. Key hyperparameters were tuned (Table 9). The final comparison on clean and corrupted data is in Table 10.

- **Hyperparameter Tuning:** Tuning revealed that $T = 1.0$, $W_{aux} = 0.4$, and $W_{kd} = 0.6$ yielded the best clean accuracy for the ResNet-50 SD model[cite: 314]. Lower temperatures consistently performed better in these experiments[cite: 313]. (See Table 9 for summary).
- **Clean Accuracy:** The optimized SD ResNet-50 achieved 88.29% accuracy, a modest +0.61 percentage point improvement over the standard ResNet-50 baseline (87.68)
- **Robustness (CIFAR-10-C):** The SD model demonstrated significantly better robustness against common corruptions[cite: 317]. It achieved an average accuracy of 74.23% on CIFAR-10-C, compared to the baseline’s 73.37
- **Analysis:** The SD mechanism, enforcing consistency between intermediate and final layers, acted as an effective regularizer, particularly boosting robustness[cite: 319, 323]. The preference for $T = 1.0$ suggests direct prediction matching was more beneficial than transferring softened "dark knowledge" in this specific SD setup[cite: 320, 326].

Table 8: Clean Accuracy Comparison (Baseline vs. Initial SD)

Method	Clean Test Accuracy (%)	Improvement vs Baseline
Baseline (ResNet-50)	87.68%	-
Self-distillation (Initial: $T=3.0$, $W_{aux}=0.4$, $W_{kd}=0.6$)	87.94%	+0.26 pp

Table 9: Hyperparameter Tuning Summary for Self-Distillation (ResNet-50 on CIFAR-10)

Temp (T)	Aux CE Weight (W_{aux})	KD Weight (W_{kd})	Test Acc (%)	Improv. vs Base (pp)
3.0	0.4	0.6	87.94	+0.26
2.0	0.4	0.6	88.23	+0.55
5.0	0.4	0.6	87.31	-0.37
3.0	0.3	0.7	87.39	-0.29
1.0	0.4	0.6	88.29	+0.61

Table 10: Robustness Test Summary (CIFAR-10-C)

Model	Clean Acc (%)	CIFAR-10-C Acc (%)	Diff. vs Baseline (%)
Baseline (ResNet-50)	87.68	73.37	-
Best SD ($T=1.0$, $W_{aux}=0.4$, $W_{kd}=0.6$)	88.29	74.23	+0.86 pp

5 Discussion

Our comprehensive experiments reveal several key insights into the workings and applications of Knowledge Distillation.

- **KD as Regularization:** Across multiple experiments (Basic KD, Re-KD, SD), a recurring theme is the role of KD as more than just knowledge transfer; it acts as a potent regularization technique[cite: 271, 278, 319, 323]. By learning from smoothed or structured label distributions (from teachers, ensembles, or itself), the student model is guided away from overfitting to hard labels and towards solutions that capture inter-class relationships and generalize better, sometimes even to unseen classes[cite: 242, 245]. This regularization effect is particularly evident in the robustness improvements seen with SD [cite: 318, 322] and the success of Re-KD[cite: 276, 278].
- **Beyond Standard KD:**
 - *Quantization:* KD synergizes well with QAT, effectively mitigating accuracy loss and enabling more aggressive quantization[cite: 255, 256].
 - *Re-KD:* The success of Re-KD challenges the necessity of a larger/stronger teacher, demonstrating that structured knowledge for regularization can flow bidirectionally[cite: 276, 279].
 - *MTKD:* Aggregating knowledge from multiple diverse teachers provides richer supervision and can lead to student performance surpassing individual teachers and even its own baseline[cite: 299, 300, 301].
 - *SD:* Teacher-free distillation is viable and particularly effective at enhancing model robustness against corruptions[cite: 318, 322, 330], likely by enforcing internal consistency[cite: 323].
- **Architectural Considerations:** Architectural match between teacher and student seems beneficial for basic KD, as seen in the MLP vs. ResNet teacher experiment on MNIST[cite: 235].

Limitations: Our study explored specific KD variants and architectures. The hyperparameter searches were not exhaustive[cite: 36, 328]. Further investigation across broader model families, datasets, and KD mechanism variations (e.g., feature-map distillation) is warranted[cite: 329].

6 Conclusion and Future Work

This project successfully validated diverse Knowledge Distillation (KD) techniques, confirming their ability to improve model accuracy, generalization, robustness against corruptions, and utility in conjunction with methods like QAT. We observed KD frequently acts as an effective regularizer via soft labels. Specifically, we verified foundational KD principles, the surprising effectiveness of Reverse KD (small teacher guiding a large student), the knowledge aggregation benefit of Multi-Teacher KD, and the robustness enhancements provided by teacher-free Self-Distillation.

Future work includes exploring alternative mechanisms (e.g., feature-map, relational), developing adaptive/hybrid methods, testing on broader architectures/tasks (e.g., Transformers), and deeper theoretical analysis including model calibration. KD remains a valuable and versatile toolkit for efficient and robust deep learning.

References

- [1] Hinton, Geoffrey, Oriol Vinyals, and Jeff Dean. "Distilling the knowledge in a neural network." *arXiv preprint arXiv:1503.02531* (2015).
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- [3] Zhang, Linfeng, et al. "Be your own teacher: Improve the performance of convolutional neural networks via self distillation." *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2019.
- [4] Polino, Antonio, Razvan Pascanu, and Dan Alistarh. "Model compression via distillation and quantization." *International Conference on Learning Representations (ICLR)*, 2018.
- [5] Yang, C., Xie, N., Wang, W., & Wang, J. (2019). "Snapshot ensemble distillation: Teacher-student optimization in one generation." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

Checklist

1. For all authors...
 - (a) Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope? **[Yes - The abstract and introduction cover the KD techniques investigated (Basic, QAT, Re-KD, MTKD, SD) and the key findings reported in the results.]**
 - (b) Did you describe the limitations of your work? **[Yes - Limitations regarding specific architectures and non-exhaustive hyperparameter search are mentioned in the Discussion section.]**
 - (c) Did you discuss any potential negative societal impacts of your work? **[No - This work focuses on foundational ML techniques (knowledge distillation) without immediate foreseeable negative societal impacts distinct from general ML model development.]**
 - (d) Have you read the ethics review guidelines and ensured that your paper conforms to them? **[Yes - Assuming authors have reviewed relevant guidelines. The work uses standard public datasets and techniques.]**
2. If you are including theoretical results...
 - (a) Did you state the full set of assumptions of all theoretical results? **[NA - The project is primarily experimental and does not present new theoretical results.]**
 - (b) Did you include complete proofs of all theoretical results? **[NA - No theoretical results requiring proofs were presented.]**
3. If you ran experiments...
 - (a) Did you include the code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL)? **[Yes - See the bottom of the page one]**
 - (b) Did you specify all the training details (e.g., data splits, hyperparameters, how they were chosen)? **[Yes - The Methodology section describes datasets (MNIST, CIFAR-10), models (MLP, ResNet), key loss functions, and specific hyperparameters (T, alpha, weights). Hyperparameter tuning procedures (grid search, optimization target) are mentioned for SD and Re-KD.]**
 - (c) Did you report error bars (e.g., with respect to the random seed after running experiments multiple times)? **[Yes - Error bars ($\pm$ std) are reported for the Reverse KD baseline and hyperparameter sweep results (Tables 5, 6). Other results should be checked if multiple runs were performed.]**

- (d) Did you include the total amount of compute and the type of resources used (e.g., type of GPUs, internal cluster, or cloud provider)? **[Yes, All experiments were run on NVIDIA L4 GPUs, consuming roughly 60 GPU-hours in total.]**
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets...
- (a) If your work uses existing assets, did you cite the creators? **[Yes - Standard datasets (MNIST, CIFAR-10, CIFAR-10-C) and model architectures (ResNet) are used. Foundational papers for KD techniques are cited in References.]**
- (b) Did you mention the license of the assets? **[Yes - Standard datasets (MNIST, CIFAR-10) and pre-trained ResNet architectures are distributed under permissive licenses that allow academic research and reuse.]**
- (c) Did you include any new assets either in the supplemental material or as a URL? **[Yes, The project's full codebase and experiment scripts are publicly available at GitHub Repository: <https://github.com/EttaZhouuuuu/ECE661Project-KD>]**
- (d) Did you discuss whether and how consent was obtained from people whose data you're using/curating? **[NA - Standard public image datasets (MNIST, CIFAR) were used, which do not require specific consent beyond acknowledging dataset terms.]**
- (e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? **[NA - Standard MNIST and CIFAR datasets do not contain PII or typically offensive content.]**
5. If you used crowdsourcing or conducted research with human subjects...
- (a) Did you include the full text of instructions given to participants and screenshots, if applicable? **[NA - No crowdsourcing or human subjects research was conducted.]**
- (b) Did you describe any potential participant risks, with links to Institutional Review Board (IRB) approvals, if applicable? **[NA - No crowdsourcing or human subjects research was conducted.]**
- (c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation? **[NA - No crowdsourcing or human subjects research was conducted.]**